

LINE2LIVE PROJECT FINAL REPORT: PIX2PIX-BASED CONDITIONAL NEURAL NETWORK FOR REALISTIC FACE PHOTO SYNTHESIS FROM SKETCH

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1 INTRODUCTION

This project intends to develop a new GAN model to revolutionize human face generation in a realistic style. In traditional art careers, creating realistic-style paintings usually requires significant time and skill. Artists often struggle to sketch suspects' facial features based on vague descriptions from witnesses, or spend massive time refining a film character to meet employers' high expectations. To address these challenges, we aim to produce a deep learning model to quickly generate realistic face images from simple sketches, benefiting not only criminal investigations but also entertainment industries like animation, game design, and personal profile pictures in social media.

With the motivation to speed up portrait generation for police sketch artists and to lower the initial cost in entertainment production, our project utilizes deep learning, specifically using U-Net and PatchGAN architecture in pix2pix model Phillip Isola (2018) as the backbones. A dataset with diverse ages, skin colour conditions and different sexes is pre-processed, along with the additional generated grayscale version of colourful target faces as intermediate generator targets. By utilizing the key advantages of the machine learning approach – faster, more powerful capability in learning complex facial patterns and continuous improvement through iterative training, our model is designed to provide significant contributions to criminal investigation and entertainment design.

2 BACKGROUND AND RELATED WORK

In recent years, numerous studies have been conducted to improve the performance of GAN for converting sketches into images. For example, the “pix2pix” model (Phillip Isola, 2018) demonstrates a straightforward solution for multiple image-to-image transformation tasks, such as converting aerial photos into realistic maps and generating shoes from sketched edges. The model architecture proposed in this paper forms the backbone of our project model architecture. Sangkloy on the other hand, resolves the general issue of poor performance in mapping the boundaries from sketches to images by integrating a CNN during the generator phase. They treat sketch-based image retrieval as a search in a learned feature embedding space, linking sketches and photos more effectively (P. Sangkloy & Hays, 2016). Chen and Hays also present a SketchyGAN model that integrates a data augmentation technique which enables complete automatic image pre-processing for over 50 categories of sketch-to-image pairs (Wengling Chen, 2018). Our team will adopt the idea of data augmentation and utilize it within our project. Moreover, Koley and colleagues explore a fresh new StyleGAN, which utilizes a “style mixing” technique where different layers of the generator are controlled by different segments of code. This solves the issue of limited user control on GAN once its architecture is established, allowing more flexibility in the GAN training phase in controlling both coarse features and finer details (Song, 2023). Another relevant investigation focused on CoGANs, two coupled GANs trained together each consisting of a generator and a discriminator. These networks are designed to learn joint distributions from samples drawn from their respective marginal distributions, which enable two GANs to generate images in different domains with a shared subset of parameters to understand the fundamental similar features between two domains. It's particularly useful when two related but different kinds of data need to be understood simultaneously, such as the colour and texture of images (Ming-Yu Liu, 2024). These previous studies established a solid foundation for knowledge in the field of converting sketches into images using GANs. In particular,

the conceptual ideas delivered by StyleGAN and CoGANs provide high-level inspiration for the project model architecture – Stacking, or task decomposition.

3 PROJECT MODEL ILLUSTRATION

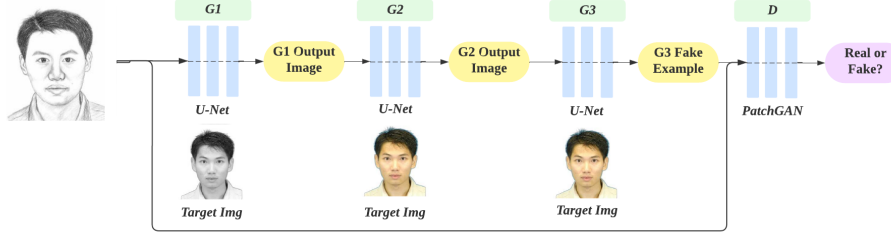


Figure 1: Project Model high-level Flow Chart

The high-level project model is a conditional GAN consisting of three generators stacked together, and the input and target images for each generator are illustrated above. If only having one generator often fails in capturing key facial features, why not assign the entire task to multiple generators, each responsible for a sub-task? In our case, we attempt to perform task decomposition into 3 parts: face structure reconstruction, basic colorization, and colour balancing. The detailed U-Net and PatchGAN architecture will be explained in the "Architecture" section.

4 DATA PROCESSING

Our team has utilized a portion of the CUHK Database from Kaggle (Zhang et al., 2011) that have complete sketch-target pair. Based on inspection, the dataset includes various Asian face types with different skin colours (light, medium, olive, brown) and ages in both sexes, providing the potential for the model to generalize on unseen data. There are a total of 2520 images including data augmentation within the dataset, with some data pre-processing steps required to complete.

First, it's important to ensure the order of reading input and target images is correct such that the model is mapping the correct sketch into faces. Unfortunately, the names of the sketches provided by the Kaggle dataset do not align perfectly with their respective faces, therefore a manual process of file renaming is completed by the team. Second, we need to resize both sketch and target images into the same 256x256 pixels with padding. It's crucial to maintain the original image ratio during resizing to prevent the model from generating faces with unrealistic proportions. Third, the background colour is also removed for target images as these unnecessary elements may potentially interfere model's learning capabilities. It may focus more on mapping the background correctly instead of focusing on face synthesis. Fourthly, since the default behaviour of the background-removed images is filling all transparent pixels with black colour, it's essential to add a white background to help the model recognize hair boundaries. Finally, the grayscale version of target images and augmented images are also generated. The choices of data augmentation are horizontal and vertical flip, rotation (90 degrees each), Five Crop (crop images into 4 corners and central crop) and Saturation (levels range from 0.5, 0.75, 1.25, 1.5), which are carefully selected to ensure the additional images added to the dataset will not negatively interfere model's ability in realistic face reconstruction (For example, random stretch is not a good choice of data augmentation as it may enable model to learn wrong size proportionality of facial features). Note that for flips, rotation, and crop type of augmentation, the same transformation has to apply on the corresponding sketch, while for saturation, the respective sketch does not require any modification but simply produces a copy of itself.

For the above total 5 steps for data pre-processing, the last 4 could be easily achieved using Python build-in module *rembg* (removebg), *PIL*, and *transforms* from *torchvision* (PyTorch). Below is a sample demonstration of data processing outputs.

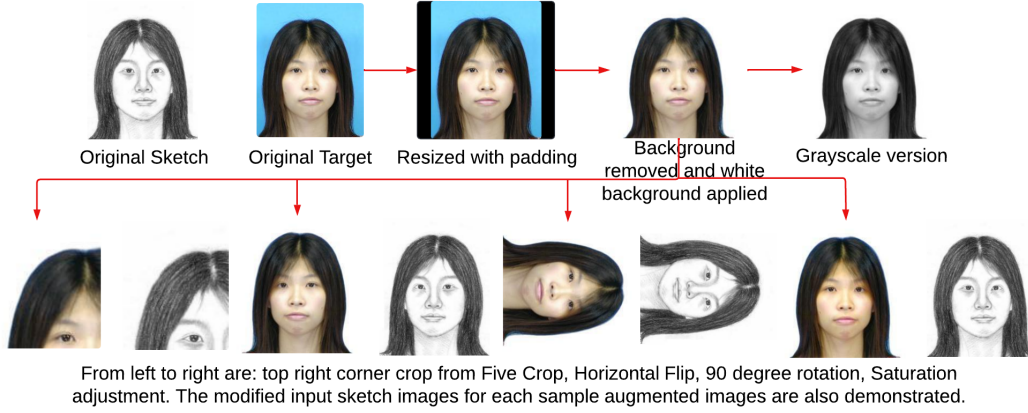


Figure 2: A Sample Demonstration of Data Processing. Note the second row only illustrates a portion of all data augmentations. The red arrow indicates the order of data processing.

5 ARCHITECTURE

All $G1$, $G2$, and $G3$ used to build the project model has the "U-Net 256" structure – modified convolutional autoencoders with skip connections to use features from multiple scales directly during the upscaling process. It consists of 8 Conv2d layers, 8 ConvTranspose2d layers, 7 LeakyReLU layers, 7 ReLU layers, 13 BatchNorm2d layers, and a final Tanh layer. The total number of trainable parameters is 54,414,531 per generator.

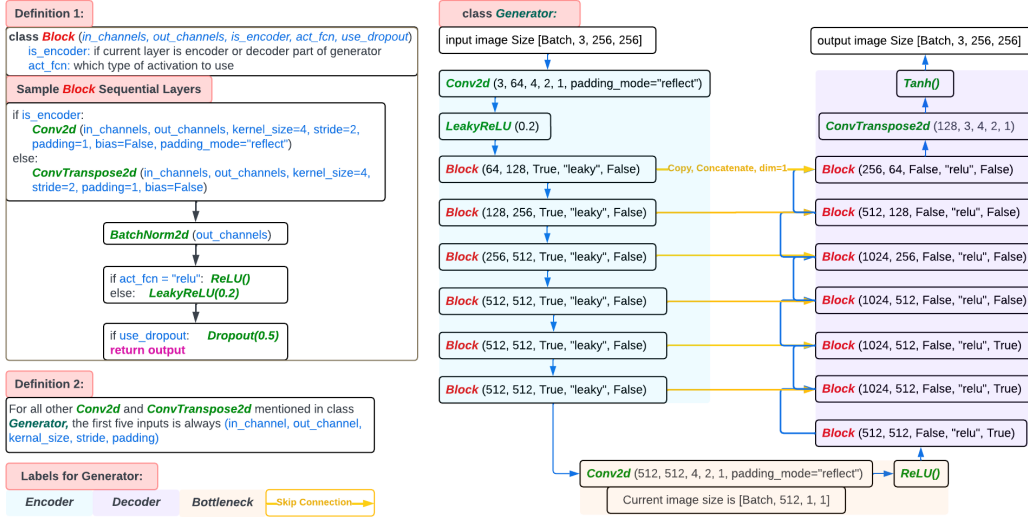


Figure 3: Project model generator architecture

The discriminator architecture used to construct the project model has the PatchGAN structure, which determines whether a portion of the $N \times N$ image is real or fake. The final prediction output for D is the average from all patches. It consists of 5 Conv2d layers, 4 LeakyReLU layers, 3 BatchNorm2d layers, and 5 GaussianNoise layers. The application of adding noise filter before each Conv2d layer is for weakening the discriminator's capability in classification (Goodfellow, 2016), which is an issue identified during the progress report that may cause the generator to receive little useful gradients from the discriminator to update itself. The total number of trainable parameters is 2,768,705.

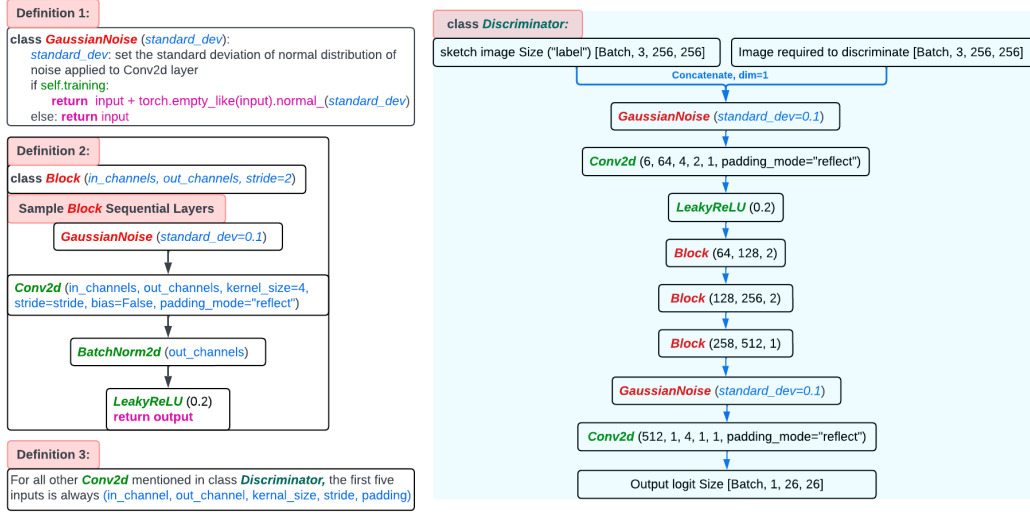


Figure 4: Project model discriminator architecture

6 BASELINE MODEL

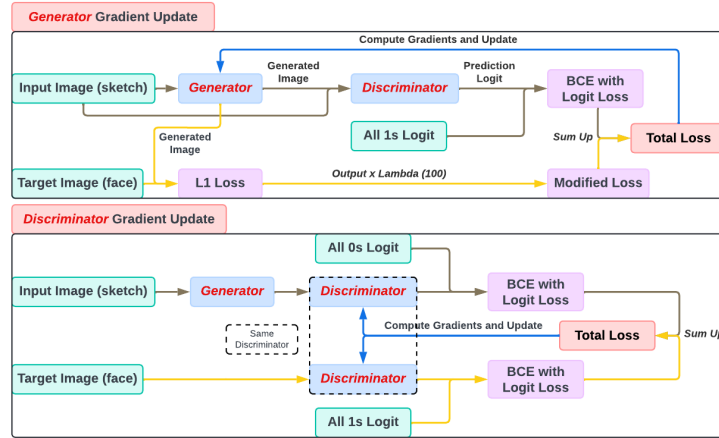


Figure 5: The logic of Generator and Discriminator Gradient Update

The baseline model is a reproduction of the pix2pix model introduced in this paper (Phillip Isola, 2018). According to the paper, it has been proved that pix2pix could accomplish and succeed in various image transformation tasks by taking input images as "conditions", such as converting aerial photos to maps. Due to the similarity of the tasks pix2pix could accomplish with our project goal, it is selected as our baseline model. An important note is that the project model has the same generator and a similar discriminator architecture (differ by Gaussian noise) as the baseline model (refer to figure 3 and 4). However, the generated images from the project model will be trained and tuned on multiple generators rather than one. Therefore, the project model would have $43 \times 3 + 12 = 141$ layers and approximately $54M \times 3 + 3M = 165$ million of parameters.

According to the paper, the generator performs best when its weight is updated with a combination of *nn.L1Loss* and the *nn.BCEwithLogitsLoss*. On the other hand, the discriminator is updated using two *nn.BCEwithLogitsLoss* to evaluate the capability to identify real images and fake images given the "condition" images (sketch in our case). Both two functions have direct implementation in Pytorch, and a detailed parameter updated procedure is demonstrated in figure 5. Note

that the same gradient updating logic applies to the project model, except we need to minimize BCE and L1 loss on all $G1$, $G2$, and $G3$.

7 QUANTITATIVE RESULTS

Finding appropriate quantitative evaluation metrics for image generation problems is always a challenging task. The famous Inception Score and Fréchet Inception Distance metrics do not apply under our project’s context, as they are both used to evaluate both realism and diversity of images. However, rewarding high diversity in the context of sketch-to-face generation means the model is generating a wide range of faces that do not necessarily resemble the input sketches, thus they both fail to measure the specific conditional relationships between the input sketch and the generated image. Therefore, some more straightforward metrics need to be introduced: Mean L1 distance, Mean L2 distance and SSIM (Structural Similarity Index).

Mean L1 distance = $\frac{1}{N} \sum_{i=1}^N |A_i - B_i|$, and Mean L2 distance = $\sqrt{\frac{1}{N} \sum_{i=1}^N (A_i - B_i)^2}$ where A_i and B_i are corresponding pixel in image A and B with N total pixels. L1 is unaffected by extreme outliers and can inform minor errors between image pixels. L2 on the other hand is more sensitive to outliers, therefore it’s useful to tell whether there are any big mismatches in major facial features. SSIM is a standard for measuring the similarity between two images based on perceived luminance, contrast and structure (edge and texture). Python scripts could be easily developed to evaluate the above metrics given above equations and SSIM has a built-in function. Below is a summary of the quantitative results evaluated on the unseen dataset. Since all the project model generators have lower L1, and L2 distances and higher SSIM than the baseline, it indicates the “task decomposition” strategy proposed by the team and the modifications to the model architecture are successful, with an improvement on L1 by 25.1%, L2 by 16.2%, and SSIM by 3.2%.

	Mean L1	Mean L2	Mean SSIM
Baseline	90.880	7.686	0.746
Project Model G1	79.730	7.252	0.767
Project Model G2	41.535	5.000	0.771
Project Model G3	83.070	7.070	0.771

Table 1: Summary of baseline and project model quantitative results on new unseen dataset

It’s commonly agreed that the loss history curve for GAN is much harder to interpret compared to the traditional CNN classification problems. Therefore, the team use this metric purely to determine whether the project model is properly trained with a consistent decreasing training loss on $G1$, $G2$, and $G3$. A validation loss of approximately 5 less than that of the baseline also justifies a better project model performance.

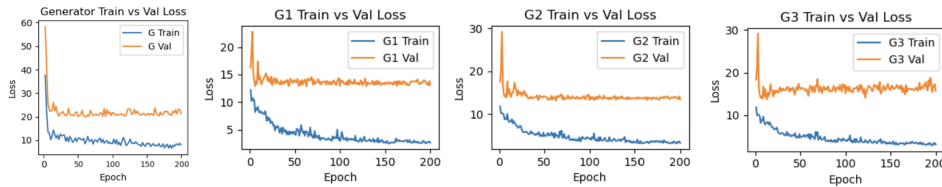


Figure 6: The History curves for Baseline G (Left 1) and Project Model $G1$ $G2$ $G3$ (Right 3)

8 QUALITATIVE RESULTS

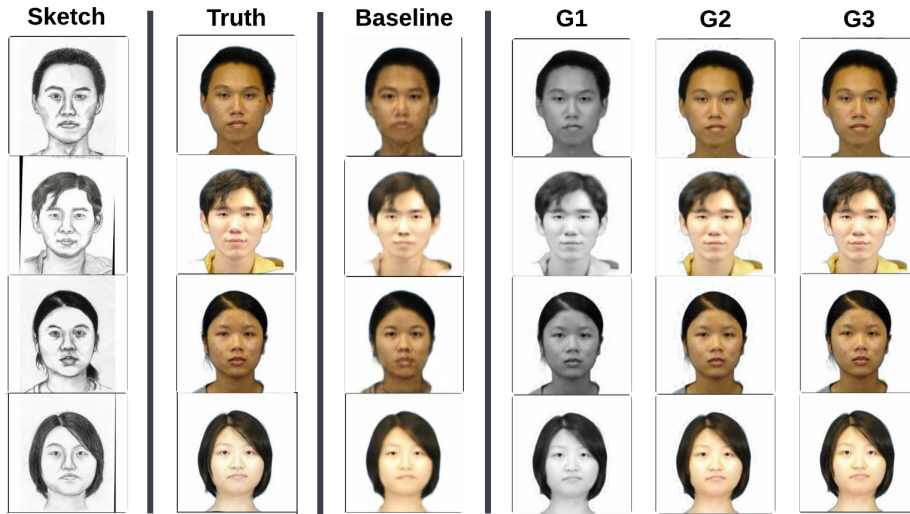


Figure 7: Sketch-truth image pairs and outputs from baseline and project model on training dataset

Comparing the outputs from $G3$ and those from the baseline, it's clear by human inspection that – 1. The project model significantly outperforms the baseline; 2. the outputs from $G3$ are almost indistinguishable from the truth images; and 3. the level of precision for image reconstruction gradually increases from $G1$ to $G3$, as one could clearly see more details of facial features, such as more refined hair and reduction in blurriness. Examining the entire outputs on the training dataset, the project model is capable of force-learning and reconstructing all inputs precisely with different sexes, ages, skin colours and face types. This should not be surprising as the model is trained on these samples.

Note that for image generation tasks, quantitative metrics should only be used as a reference for evaluating models when a clear conclusion cannot be reached through simple human inspection. The deterministic factor is still human perception (Do we believe, and perceive the generated images as high-quality?). It's biased to claim the model with the least L1/L2 distance and/or highest SSIM is the best model since these evaluation metrics only reflect one certain aspect of the generation quality. Therefore, the team will not connect qualitative results to quantitative metrics, as the differences in outputs by each generator are obvious to human raw eyes.

9 EVALUATE MODEL ON NEW DATA

The original dataset was divided into 90% of training, 5% of validation and 5% of testing using a stratified sampling approach to ensure the testing dataset cover a wide variety of Asian faces that contain different sex, ages and skin colour. This dataset was reserved and only exposed to the model until writing this report to validate the model's effectiveness under realistic conditions. Due to the poor drawing techniques all team members have, and the time investment required to learn realistic sketching skills, the approach to mimic the original dataset sketch style and create new testing samples is not feasible given the context of this project. The quantitative results on the new dataset are already demonstrated in and discussed in table 1. Qualitatively, outputs on unseen datasets from baseline and $G1$, $G2$, $G3$ are illustrated in fig. 8.

Although some of the reconstructed images may deviate slightly from the original, the team intentionally selected these images to encourage interesting observations and discussions in a later section. We aim to present a balanced view, including both the successful and less successful examples, rather than only showcasing the best results. Similar to the examples illustrated on the training dataset, it's evident that the outputs from the project model on the unseen dataset outperform the

baseline – baseline images are more blurry, contain less sharp facial feature details, have distortions on facial symmetries and proportions, reveal visible artifacts around edges of faces, and have more unnatural colouring compared to $G3$ images. Therefore, from both qualitative insights and quantitative results (fig. 8 and table 1), the team can confidently claim that the project model surpasses the performance of the existing pix2pix baseline model.

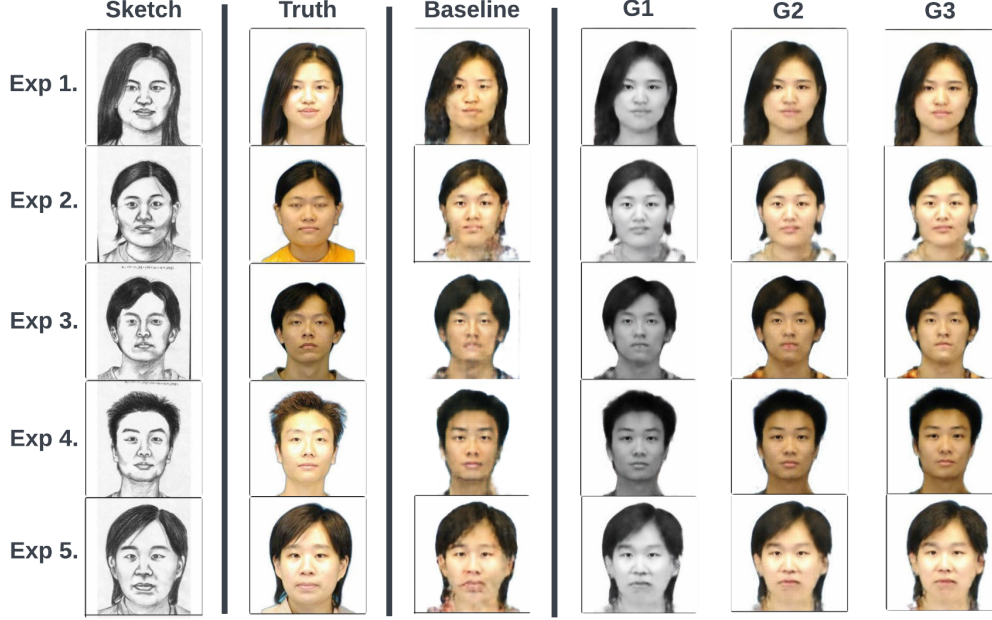


Figure 8: Sketch-truth image pairs and outputs from baseline and project model on unseen dataset

10 DISCUSSION

The first interesting observation is derived from the analysis on specific values in table 1. The extremely low L2 values combined with high L1 values across all four generators suggest that while there are numerous pixel-level differences in the reconstructions, these differences are minor enough not to result in large squared errors. This indicates that, overall, the generated images closely resemble real images at a higher level. The relatively high SSIM values (with 1 being the maximum) further confirm that the generated images maintain strong similarities with truth images, particularly in terms of face shape, textures, and contrast. The slight improvement in SSIM (3.2%), a moderate improvement in mean L2 (16.2%), and significant refinement in L1 from baseline (25.1%) also suggest that the project model has undergone small enhancement in structural integrity but noticeable optimization in pixel-level accuracy, resulting in much more natural face synthesis with correct and precise facial features location and colouring.

One might argue that based on the quantitative results in table 1, $G2$ outperforms $G3$ and thus only the first two generators should be retained. However, a closer examination of $G2$'s outputs reveals subtle but consistent "pixelated grid-like" patterns and black dots around the borders of the faces. These "grid-like" patterns are even more pronounced in $G1$'s outputs. In contrast, $G3$ effectively eliminates the black dots and significantly reduces the "grid-like" patterns, resulting in more natural-looking reconstructions. The cause of such "grid-like" patterns is due to the GaussianNoise layer added to the discriminator. While it diminishes the discriminator's power and allows for more valid gradient updates on G , it can also cause G to learn unusual noise patterns that deceive the D . Therefore, including $G3$ is crucial for enhancing the qualitative results. This also supports the earlier statement that, in this project's context, the quality of the generated outputs is much more important than the quantitative metrics alone.

The second noteworthy observation is the model’s ability to handle unnatural face shapes. If we examine example 2 in fig. 8, we would agree it’s a difficult face to generate as the hollow cheek feature is not commonly found in Asian facial types. Moreover, our training dataset lacks any face structure as unique as the one in example 2. Surprisingly, the project model successfully reproduces this face with a high degree of similarity to the original image, which demonstrates the model’s robustness and ability to generalize even to challenging and uncommon facial features.

Another observation result is the model’s inconsistency in reproducing skin tones. In examples 2, 3, and 4, we can observe noticeable skin colour mismatches between the *G3* outputs and the ground truth, despite the model’s near-perfect replication of the facial features’ shape and location. During training, the team tried to teach the model to recognize multiple skin tones by applying various levels of saturation to ground truth images. However, we miss the fact that the input sketches, the “conditions”, provided no information about skin colour at all (they are purely grayscale). As a result, even though the model is likely to learn multiple colour schemes based on the different saturation levels, it tends to apply one somewhat arbitrarily during generation, which causes the observed mismatches in skin colour. Fortunately, in terms of practical applications such as identifying a person from a portrait, the accuracy of facial feature location and shape is much more critical compared to skin color. Skin colour can vary significantly due to lighting conditions and exposure, whereas the underlying structure of facial features remains consistent and reliable for identification purposes.

The final and most intriguing observation is that the model’s output is heavily influenced by the type of lines used in the input sketches. In examples 1-4, where the *G3* outputs show clear facial borders, it’s evident that the artist emphasized the face borders in the original sketches by using hard, black contour lines, in contrast to the softer shading lines used to indicate facial wrinkles. However, in example 5, the contour line at the bottom of the face in the sketch is not as “hard” and black, leading to a failure in reconstructing the face border in the *G3* output (the neck and bottom face somewhat merge together). Additionally, in example 3, the face border in the input sketch does not accurately represent the ground truth image – the actual face has one side of the chin higher than the other, but the input sketch represents them at the same level. As a result, the *G3* output closely follows the input sketch with a “flat” chin, leading to a slight mismatch in the high-level face structure. These two examples indicate that while the model demonstrates strong generalizability and performance on unseen datasets, the quality of the input sketches with clear and well-defined lines plays a critical role in producing optimal final output.

Throughout this project, we’ve gained valuable insights into completing a complex machine-learning task when working with large-scale models with additional constraints as inputs. We build an intrinsic understanding of how each hyperparameter on every aspect of the training scripts, including layers in architecture, the choice of loss functions to minimize, the type of data augmentation to perform, as well as the impact of input sketch images on the final outputs. This knowledge enabled us to effectively analyze and modify the project model to optimize its performance. We’ve encountered challenges in reconstructing the training loop for establishing the stacked generator framework, but we ultimately gained enough experience in PyTorch to implement the desired project model structure through trial and error. We’re equipped with advanced techniques and a deeper understanding in the field of image generation, which will support us in tackling future machine-learning challenges.

11 ETHICAL CONSIDERATIONS

One of the primary ethical considerations arises from the context of the project itself. In order to train the GAN model, a huge amount of training data with sketch-selfies pair is required. However, portrait rights are a sensitive issue in today’s society, especially with the growing reliance on big data. Although the data sources used for the training model have obtained permissions, a massive number of additional training samples must be collected to further improve the generalizability of the model. This will raise significant privacy concerns on the legitimacy of the store and use of data, which requires careful attention to data transparency.

Another concern is the inherent bias in the original dataset. Although the dataset is diverse in terms of age and sex, it only contains images of Asian faces. As a result, the model is likely to generate realistic faces with Asian facial features from any input sketch. This is particularly critical in the context of facial composition for criminal investigations. The model’s bias could potentially influence police decision-making, leading to wrongful accusations. The team recognizes this limitation

and believes that while there's no existing database that includes Western faces with artistic sketches in pairs, taking steps to start building this dataset is essential to mitigate biases.

Finally, the model could also potentially be used to create images for fraudulent purposes or to impersonate others, causing both financial and credit risks. To prevent this false representation to individuals, developing comprehensive policies and constructing a clear accountability structure are required to track the use of generated images.

12 PROJECT DIFFICULTY

Completing the project of generating realistic face images from input sketch has several significant challenges – the team has zero knowledge on how to train multiple generators with different target inputs simultaneously, the massive GPU demand and longer training times, as well as the extreme difficulty in making decisions for hyper-parameter adjustments.

First, since there's no lab related to image generation with an input condition, the team has little context for constructing the basic coding blocks for training the networks. Therefore, the team initially spent massive time understanding the U-Net and PatchGAN architecture, researching how to insert a "label" into the generator, learning to create custom datasets to load sketches, target face images and the grayscale version of face images simultaneously, as well as studying possible training loop structure for training 4 networks together (3 G and 1 D). The absence of clear instructions for completing these tasks, particularly the mandatory reconstruction of the training loop function compared to the one used in the lab, led to massive trial-and-error efforts.

The second issue was the significant time and GPU resources demanded to train our project models. As we are essentially training 4 networks within one trial, the computational cost is at least 4 times greater compared to the course lab. To resolve this challenge, the team researched and implemented the Automatic Mixed Precision package from PyTorch, which allows customization of the precision floating point numbers of the inputs during training. By utilizing the 16-bit precision for the computing environment, the team managed to train with larger batch sizes, reduced GPU memory usage and faster training time.

The inherent nature of the image generation task, with a combination of the high computational cost of our project, introduces the greatest challenge – decisions on hyper-parameter tuning. Unlike our course labs where evaluation is straightforward with clear truth labels, the team spend a lot of time simply selecting and justifying the suitable quantitative metrics. We reviewed 15+ papers related to GAN implementation and analyzed the most frequently used metrics. Then we analyzed each metric theoretically and discussed whether they matched the context of our project (conditional generation + reconstruction accuracy focus, diversity excluded). However, the team understand that these quantitative metrics are just a guideline. It's inappropriate to claim the model is THE best with the least L1/L2 losses as each metric only measures one aspect of the generated images. What matters the most is always the perceptual quality – do we, human observers perceive the generated image as high-quality? With this agreement, we can now easily see the challenge of the hyper-parameter tuning for this project. The team have experimented with different combinations of loss functions minimized on generators (for example, BCE on first G only, on second G only, on third G only), added both DropOut and Gaussian noise on D layers, changed the target image type corresponding to each G (for example, all G have colourful images as target) and so on. The team took careful consideration for every modification to the model training logic given such high computational costs. Nevertheless, we must rely on subjective human judgment and the limited guidance from quantitative metrics to evaluate whether our tuning efforts are successful.

In conclusion, by completing this project, the team not only honed advanced PyTorch coding skills for training complex models but also gained valuable experience in decision-making under constrained conditions. This involved logical reasoning and analysis, alongside the use of reference metrics to guide our judgements, which ultimately enhanced our problem-solving capabilities in machine learning applications.

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